

PRANVEER SINGH INSTITUTE OF TECHNOLOGY

Major Project Proposal

Team Id: 27_CSAI_4A_09

Team Details:

S.No.	Full Name	Roll No	Branch & Section	Mob No
1	Bhavya Gupta	2301641520064	CS AI 3-A	9569461013
2	Bhumika Mishra	2301641520065	CS AI 3-A	9336979743
3	Divyansh Gaur	2301641520070	CS AI 3-A	9026560306
4	Divyanshi Singh	2301641520072	CS AI 3-A	9548482611

Project Title:

Domain:

1. Software-Web Application	2. Agentic AI
3. Artificial Intelligence/Machine Learning/Deep Learning/Gen AI/Agentic AI	4. Cyber Security

Problem Statement:

Modern software development heavily depends on open-source libraries, distributed repositories, and rapid deployment practices. While this accelerates innovation, it also introduces significant security risks such as insecure coding practices, vulnerable dependencies, and misconfigured containers. Developers often rely on multiple security tools to detect vulnerabilities, but these tools generate fragmented results that are difficult to analyze and prioritize. As a result, development teams struggle to identify critical vulnerabilities quickly and take appropriate action. Additionally, many developers lack the expertise to interpret security reports and understand the impact of detected vulnerabilities. This creates a gap between vulnerability detection and effective remediation.

Therefore, there is a growing need for an intelligent and unified platform that can analyze repositories, detect security issues across multiple layers, prioritize risks, and provide actionable insights to help developers build secure and reliable applications.

Proposed Solution:

The proposed system is an AI-Powered Unified Vulnerability Intelligence Platform designed to identify and analyze security vulnerabilities in software repositories. The platform allows users to initiate security scans through multiple entry points such as GitHub OAuth authentication, manual repository URL input, or a VS Code extension. Once a scan request is received, the backend API creates a scan job which is processed through a Redis-based job queue and executed in an isolated Docker container to ensure system security. The system integrates established security tools such as Semgrep for static code analysis and Trivy for dependency and container vulnerability scanning.

The outputs generated by different scanning tools are then normalized into a unified vulnerability format and processed by a Risk and Heat Modeling Engine that calculates risk scores and file-level heat scores. These results are presented through an interactive web dashboard that displays vulnerability details, risk meters, and heat maps to highlight critical areas in the codebase. Additionally, an AI-based module provides developers with explanations of vulnerabilities and recommended secure coding fixes, enabling faster and more informed remediation of security issues.

Unique/Distinctive feature of the solution:

1.Explain the uniqueness and distinctive features of the solution

The proposed system offers a unified vulnerability intelligence platform that integrates multiple security scanning tools into a single framework for analyzing application code, dependencies, and container configurations. Unlike traditional tools that produce fragmented reports, the system introduces a Risk and Heat Modeling Engine that converts raw vulnerability data into structured risk scores and visual heat maps, helping developers easily identify critical security issues.

The platform also includes a VS Code extension for direct project scanning and an AI-based assistant that provides explanations and secure remediation suggestions. These features make the solution more efficient, developer-friendly, and capable of improving overall software security.

2.How your proposed solution is different from similar kind of products by the competitors

Most existing security tools focus on a single layer of analysis, such as static code scanning or dependency vulnerability detection, and often generate fragmented reports that are difficult for developers to interpret. In contrast, the proposed platform provides a unified vulnerability intelligence system that integrates multiple security engines and normalizes their outputs into a structured format. The system further introduces a Risk and Heat Modeling Engine that prioritizes vulnerabilities through risk scores and visual heat maps.

PRANVEER SINGH INSTITUTE OF TECHNOLOGY

Additionally, the integration of a VS Code extension and AI-based remediation assistance helps developers understand vulnerabilities and apply fixes directly within their development workflow, making the solution more practical and developer-centric.

Tools/Technology Uses:

1) Hardware Requirements

- a) Processor: Intel Core i5 / Ryzen 5 or higher
- b) RAM: Minimum 8 GB (16 GB recommended for running Docker containers)
- c) Storage: Minimum 256 GB SSD
- d) Graphics Card: Not required (Integrated graphics sufficient)
- e) Others: Stable internet connection for repository access and dependency scanning

2) Software Requirements

- a) Operating System: Windows 10/11, Linux, or macOS
- b) Development Tools: Visual Studio Code, Node.js, Docker, Git
- c) Database: PostgreSQL
- d) Web Browser: Google Chrome / Microsoft Edge / Firefox
- e) Others: Redis (for job queue), Sengrep (SAST scanning), Trivy (dependency & container vulnerability scanning), GitHub API for repository access

3) Others (If any)

React.js / Next.js for frontend dashboard, Tailwind CSS or shaden UI for interface design, Recharts or Chart libraries for visualization, and AI API integration for vulnerability explanation and remediation suggestions.

PRANVEER SINGH INSTITUTE OF TECHNOLOGY

(To be Filled by Faculty/Evaluator)

Proposal Evaluation:

1. Right Identification of the Problem (Appropriate selection of the problem)?
a) Excellent b) Good c) Needs Improvement d) Unacceptable
2. Relevance of the Solution (Adequately addressing the problem/need)?
a) Excellent b) Good c) Needs Improvement d) Unacceptable
3. Innovativeness in the Solution (Distinctive innovative components/features of the solution)?
a) Excellent b) Good c) Needs Improvement d) Unacceptable
4. Uniqueness of the Solution (Intellectual Property Component)?
a) Excellent b) Good c) Needs Improvement d) Unacceptable

Improvements/ Suggestions by the Evaluator:

1.
2.
3.
4.

Name of Faculty In-charge:

Designation:

Signature with Date:

Guidelines:

- One Proposal per team will be submitted by the team leader only.
- A Team can have maximum 5 Members.
- For Heading font Size=14, Content font size =12, single spacing, Times New Roman type only.